

Data Predictions and Biostatistics

JOHANNES KEPLER UNIVERSITÄT



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BUT 2 Science des Données



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Foreword

When I started studying for my Data Science degree, I had one main goal: to do an internship abroad. I've always valued cultural openness and the chance to step out of my usual environment. Doing an international internship seemed like a unique opportunity to combine academic learning, personal growth, and the discovery of a new way of life.

This internship at Johannes Kepler University Linz, in Austria, was much more than just a step in my studies—it was a turning point in my journey. Working in a well-known statistics department, being part of a multicultural environment, and living in a city I didn't know helped me become more independent, more adaptable, and gave me a better understanding of how science is approached in different parts of Europe.

This report shows everything I learned during this experience, both technically and personally. It also represents the achievement of a goal I had in mind since the beginning of my studies.

Table of Contents

<i>Host Institution Overview</i>	3
I Johannes Kepler Universität Linz.....	3
II The Department of Applied Statistics	4
III My Integration Into the Team	4
<i>Main Assignments</i>	5
I First Project.....	5
II Second Project	6
<i>Main Project 1: Electricity Cosumption Forecasting for Linz AG</i>	7
I Project Context and Objectives.....	7
II Data Preparation	8
III Exploratory Analysis	8
IV Modeling	11
V Conclusion.....	18
<i>Main Project 2: Biostatistical Study on Yeast DNA Evolution</i>	19
I Introduction	19
II Understanding the Study.....	20
III My Work.....	21
IV My Work.....	21
<i>Additional Activities</i>	24
I Attending Courses in German and English.....	24
II Seminar	24
III Language and Cultural Experience	25
<i>Final Reflection</i>	26
I Professional Experience	26
II Personal Experience.....	27
III Final Conclusion	27
<i>Glossary</i>	28
<i>Appendix</i>	29

Host Institution Overview

I – Johannes Kepler Universität Linz



Figure 1 Around the lake of JKU

Located in Upper Austria, Johannes Kepler University Linz (JKU) is a modern and dynamic public university founded in 1966. It is named after the famous astronomer and mathematician Johannes Kepler, who lived in Linz in the 17th century. Today, the university is known for its academic excellence, interdisciplinary research, and international outlook.

The main campus, in the north of Linz, offers a pleasant and green study environment. It includes four main faculties: Social Sciences, Economics and Law; Law; Medicine; and Engineering and Natural Sciences. In recent years, the university has also created a School of Education and the Institute of Digital Sciences Austria (IDSA), which focuses on digital technologies and artificial intelligence.

JKU provides a stimulating environment for internships, research projects, and academic exchanges. The Department of Statistics and Data Science, where I did my internship, plays a key role in teaching applied data analysis, both in theory and in practice.



Figure 2 Around the lake of JKU



Figure 3 Forest next to Science Park

II – The Department of Applied Statistics



Figure 4 JKU Science Park

The Department of Applied Statistics is located in the Science Park 2 building at JKU.

It includes three university professors, four administrative staff, nine research associates, five PhD students, and seven student assistants.

The Institute of Applied Statistics at JKU Linz offers a complete academic program in statistics and data science. The *Bachelor in Statistik und Data Science*, taught in German, introduces students to the basics of data collection, analysis, and visualization. The *Master in Statistics and Data Science*, taught in English, allows students to deepen their

knowledge of statistical methods. Finally, the Institute also offers a PhD program for those who want to pursue research in applied statistics. These programs combine theory, practice, and an international perspective.

III – My Integration into the Host Institution

I integrated into the university in several ways. I had the chance to meet a few Master's students while occasionally working in the same room as them. But I spent most of my time having lunch with PhD students, professors, and assistants.

I also had the opportunity to visit a botanical garden in Linz with my supervisor and other members of the department. Afterwards, we went to a traditional restaurant in Linz, where I discovered Wiener Schnitzel and asparagus soup. These moments helped me connect with my colleagues and also allowed me to experience Austrian culture.



Figure 5 Visiting the botanic garden

Résumé de mes deux missions principales (Français)

Au cours de mon stage au sein du département de statistiques de la Johannes Kepler Universität de Linz, j'ai eu l'opportunité de travailler sur deux projets principaux, à la fois complémentaires et formateurs.

I - Prédiction de la consommation d'électricité de la Linz AG

Ce projet s'inscrivait dans le cadre du concours *Linz AG Prognosechallenge 2024/2025*, dont l'objectif était de concevoir un modèle de prévision de la consommation électrique à court terme. Après avoir fusionné des données de consommation (toutes les 15 minutes) avec des données météorologiques issues de plusieurs localités, j'ai mené une analyse exploratoire approfondie pour identifier les effets saisonniers, journaliers et hebdomadaires sur la demande en électricité.

La modélisation a d'abord reposé sur des régressions linéaires multiples, puis s'est orientée vers des approches de machine learning plus performantes : **Random Forest*** et **XGBoost***. Ces méthodes ont permis d'atteindre un **R^2 supérieur à 0.85** lors des prédictions sur données futures. Le projet m'a permis de mettre en œuvre des compétences avancées en data science, à la fois en R et en Python, et d'aborder des enjeux concrets liés à la transition énergétique.

II - Étude bio-statistique sur l'évolution de l'ADN des levures

Dans un second temps, j'ai participé à la reproduction des résultats graphiques d'une étude scientifique publiée en 2021 (*Phillips et al.*), portant sur l'évolution génétique de populations synthétiques de levures. L'objectif était de comparer deux méthodes de croisement (aléatoire et structuré) en termes de conservation de la diversité génétique au fil des générations.

Après avoir étudié l'article en détail, j'ai développé, en langage R, des scripts permettant de visualiser les **fréquences alléliques*** à différents cycles de reproduction. Ce travail m'a permis de renforcer mes compétences en biostatistiques, en analyse de données génétiques, et en communication scientifique.

Summary of My Two Main Projects

During my internship at the Department of Statistics at Johannes Kepler University Linz, I had the opportunity to work on two major and complementary projects, which allowed me to apply and further develop my skills in data science and statistical analysis.

1 - Electricity Consumption Forecasting for Linz AG

This project was part of the *Linz AG Prognosechallenge 2024/2025*, which aimed to develop a short-term electricity consumption forecasting model. I first merged high-frequency electricity consumption data (15-minute intervals) with meteorological data from multiple locations in Upper Austria. A thorough exploratory analysis helped identify seasonal, weekly, and daily patterns in energy demand.

The modeling phase started with multiple linear regression and progressed to more advanced machine learning approaches, including **Random Forest** and **XGBoost**. These models achieved high predictive performance, with **R² values above 0.85** when tested on future data. This project allowed me to apply a wide range of data science techniques using both R and Python, while deepening my understanding of energy-related forecasting challenges.

2 - Biostatistical Study on Yeast DNA Evolution

My second assignment involved reproducing graphical results from a scientific paper published in 2021 (*Phillips et al.*), which investigated how different crossing strategies affect genetic diversity in synthetic yeast populations. The study compared two methods: a simple mixing approach and a more structured pairwise crossing strategy.

After analyzing the study in detail, I wrote R scripts to recreate key visualizations, such as allele frequency spectra across several evolutionary cycles. This work strengthened my skills in biostatistics, data visualization, and interpretation of genetic data in a controlled experimental context.

First Project: Forecasting Electricity Consumption for Linz AG

I – Project Context and Objectives

As part of the “Linz AG Prognosechallenge 24/25,” the goal was to develop a short-term electricity consumption forecasting model. This project takes place in an energy context where accurate forecasts are essential to maintain the balance between electricity production and consumption. Indeed, a mismatch between predicted and actual consumption can lead to significant extra costs, especially due to redispatching measures.

Electricity consumption is measured and billed in 15-minute intervals, and it is strongly influenced by time-related factors such as the day of the week, holidays, or the season. With the rise of photovoltaic (PV) systems and heat pumps, electricity usage has become even more sensitive to weather conditions, adding complexity to forecasting. Moreover, the non-stationary nature of the process—due to changing customer behavior—makes the task even more challenging.

The available data cover the period from January 1, 2021, to May 1, 2024, for electricity consumption, and up to November 1, 2024, for weather data and standard load profiles (SLP). These profiles describe typical consumption patterns for different types of users (households, businesses, etc.) and can help improve the forecasts. The weather variables come from six locations in Upper Austria and include temperature, solar radiation, wind speed, and precipitation.

The final objective is to accurately predict electricity consumption between May 1 and November 1, 2024, using all available data sources. Model performance will be evaluated using the R^2 (coefficient of determination).

II – Data Preparation

The data preparation phase was a crucial step in the project. Its goal was to structure and enrich the raw data to make it usable for statistical modeling. Two datasets were first imported: the aggregated electricity consumption data (*gesamtabsatz*), measured every 15 minutes, and the weather data collected from six different locations. These datasets were merged using the timestamp to create a combined dataset.

The weather data were originally organized by city and by variable (e.g., *temp_Linz*, *temp_Gramastetten*). To reduce dimensionality while keeping relevant climate information, an average was calculated for each type of weather variable (temperature, wind, precipitation, etc.) across all locations for each time interval. This created a final dataset (*data_avg*) that includes, for each time step, the electricity consumption and the averaged weather variables.

This dataset served as the foundation for the rest of the analysis. It was cleaned of missing values and enriched with derived time-related variables (such as the season). It was then exported as a CSV file to ensure it could be reused later for modeling or exploratory analysis.

III – Exploratory Analysis

Exploratory analysis is a key step in any predictive modeling project. In the context of the Linz AG Prognosechallenge 2024/2025, this phase aimed to identify seasonal, weekly, and daily patterns in electricity consumption, as well as weather-related factors that might explain its variation. The analysis was based on two datasets: electricity consumption recorded every 15 minutes from January 2021 to May 2024, and aggregated weather data from six locations in Upper Austria covering the same period.

1. Analysis of Averaged Weather Data

The first step was to group weather variables by type (temperature, wind, solar radiation, precipitation) by calculating their average across all locations. This simplification helps reduce dataset complexity while keeping useful regional weather information for forecasting.

To assess the impact of each weather variable on electricity consumption (*gesamtabsatz*), simple linear regressions were performed, each using one averaged weather variable as a predictor. The results show that average temperature is the most correlated with consumption, with an R^2 of 0.111. Wind speed comes next ($R^2 = 0.053$), followed by direct solar radiation ($R^2 = 0.031$). Precipitation has almost no explanatory power ($R^2 < 0.001$). Although these values are modest, they confirm that weather conditions affect energy demand, especially for heating and cooling needs, but they are not enough alone to model the full complexity of electricity consumption.

2. Descriptive Statistics and Seasonality

Descriptive statistics for the *gesamtabsatz* variable show an average electricity consumption of 122.5 MW, with a standard deviation of 30.13 MW. The distribution is relatively wide, with a first quartile at 99 MW, a median at 119.3 MW, and a third quartile at 143.3 MW. Monthly analysis reveals clear seasonality: the highest values occur in winter (145 MW in December, 144 MW in January), while the lowest are seen during the summer months (around 106 MW in June, July, and August). These results reflect higher energy needs for heating in winter, which is typical in European countries with a continental climate.

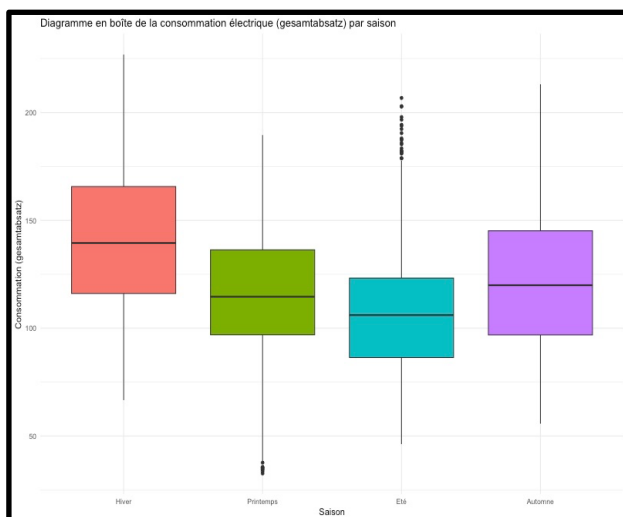


Figure 6 Boxplot of the seasonality of the electric consumption

This seasonality is also visible in the boxplots, which show lower variability during the summer and a concentration of high consumption values in winter. A decomposition of the daily time series using an additive model confirms the presence of a strong seasonal component, along with a slightly decreasing trend over the observed period. This trend may be explained by improvements in energy efficiency or an increase in photovoltaic production, which helps reduce net electricity consumption.

3. Weekly and Daily Effects

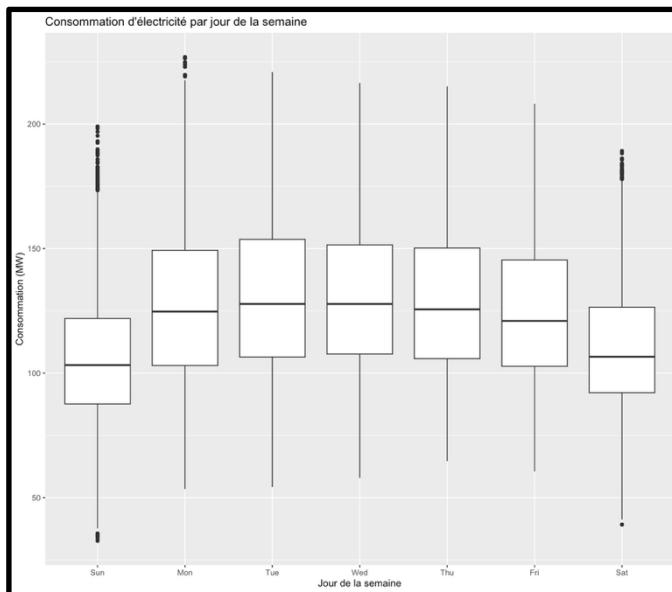


Figure 7 Boxplots of the weekly pattern of the electricity consumption

Electricity consumption also shows weekly patterns. On average, it peaks on Tuesdays and Wednesdays (around 130 MW) and drops significantly on weekends, reaching a minimum on Sundays (around 107 MW). This pattern matches the typical rhythm of industrial and service-sector activity, which is higher during weekdays.

Daily effects are even more pronounced. When averaging consumption by hour of the day, a typical “double peak” curve appears: after a low point around 3 a.m. (about 91 MW), consumption rises sharply to a peak at 7–8 a.m. (around 140 MW), then slightly decreases during the day before rising again in the early evening. This pattern reflects human activity cycles, such as the start of the workday, heating or cooling needs, and occupancy in homes.

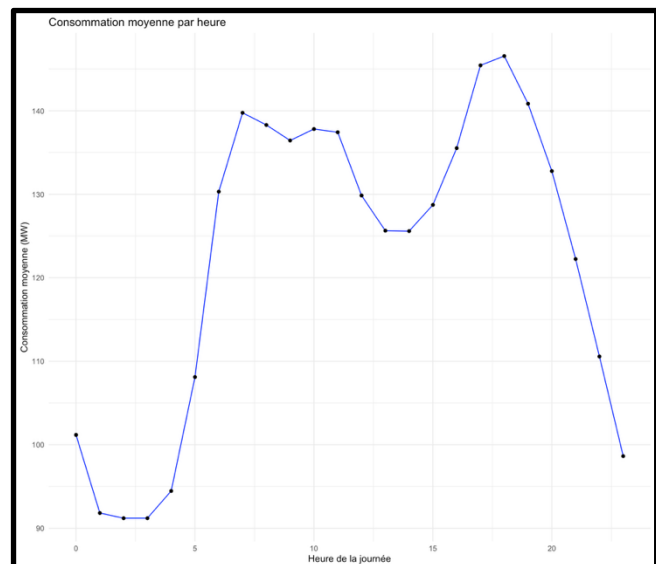


Figure 8 electric consumption in a average day

Statistical correlations confirm these observations. The Pearson correlation between hour of the day and consumption is estimated at 0.279 ($p < 2.2e-16$), while the Spearman correlation—more robust to non-linear relationships—reaches 0.306. These values suggest a significant monotonic relationship between the two variables, highlighting a clear daily effect.

Conclusion of the Exploratory Analysis

The exploratory analysis showed a moderate influence of average temperature on electricity consumption, while other weather variables—such as wind, solar radiation, and precipitation—had a more limited impact. Consumption showed strong seasonality, with peaks in winter and more stable levels in summer, along with weekly and daily variations linked to patterns of human activity.

These observations highlight the value of using multiple linear regression to model electricity consumption by combining both weather-related and time-based factors, in order to improve prediction accuracy.

IV – Modeling

1. Multiple Linear Regression with Weather Data

Our first step in building a model to predict electricity consumption was to use a Multiple Linear Regression (MLR) model based only on daily average weather variables. The model was fitted using the following formula:

```
modele <- lm(gesamtabsatz ~ dhi_mean + dni_mean + ghi_mean +  
              precipitation_mean + temp_mean + windspeed_mean,  
              data = data)
```

Figure 9 RLM model with R

The model is statistically significant overall ($F = 5185$, $p < 2.2e-16$) with an adjusted R^2 of 0.21, meaning that 21% of the variability in electricity consumption is explained by the included weather variables.

To improve the model, we decided to add categorical time variables.

2. Multiple Linear Regression with Weather and Time Variables

To improve the explanatory power of the model, we extended the previous Multiple Linear Regression by including time-related variables: hour of the day, day of the week, and month. The model is defined as follows:

```
modele <- lm(gesamtabatz ~ dhi_mean + dni_mean + ghi_mean +  
             temp_mean + windspeed_mean +  
             heure + jour_semaine + mois,  
             data = data)
```

Figure 10 RLM model with R

The model shows an adjusted R^2 of 0.4461, meaning that nearly 45% of the variability in electricity consumption is explained by the combined weather and time-related variables. This represents a significant improvement compared to the model based on weather data alone ($R^2 \approx 21\%$).

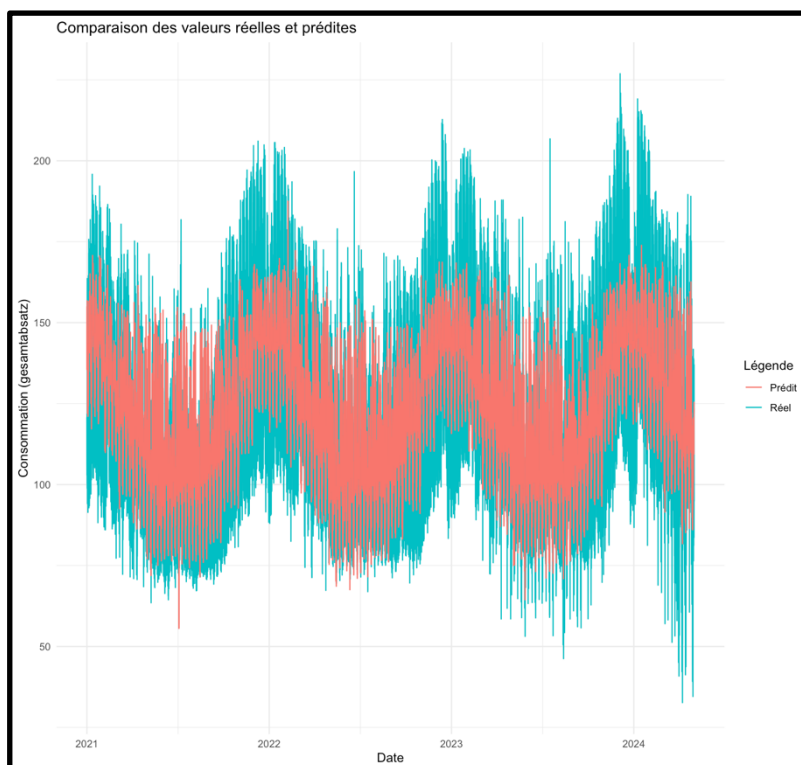


Figure 11 Comparison between prediction and reel values with RLM

However, the prediction accuracy is still not sufficient, even when using all available variables. Multiple Linear Regression has its limits, and to achieve better results, we need to consider a more advanced method — Machine Learning.

3. When in doubt, use Random Forest !

A Random Forest is a supervised machine learning algorithm used for both classification and regression tasks. It is an ensemble method that combines multiple decision trees trained on different subsets of the data, introducing randomness in the selection of variables at each split. The goal is to reduce overfitting and improve the model's accuracy.

The first step was to train the model on the full dataset. The R^2 between the actual and predicted values reached 0.98, showing that the model almost perfectly reproduces the observed data. We can therefore use this model to forecast future electricity consumption, relying on both categorical time variables and weather forecasts.

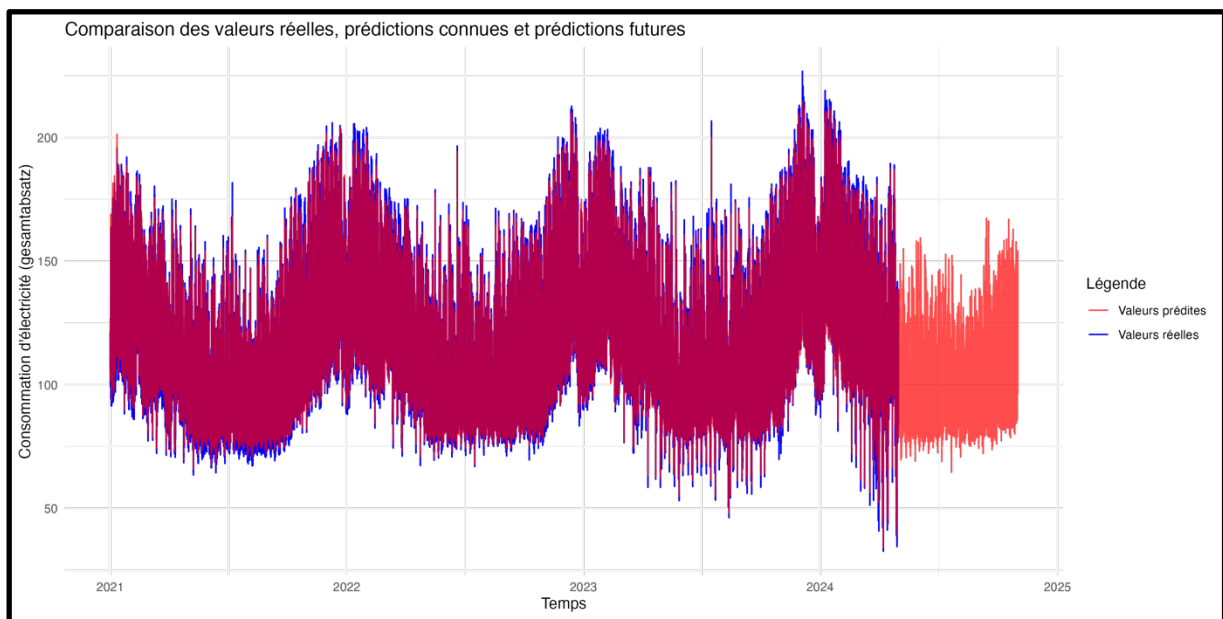


Figure 12 Comparison between prediction and reel values with Rnadam Forest

To validate the quality of these predictions, we needed to perform additional tests.

The experiment we conducted was as follows: we trained the model on data from 2021, then used it to predict values for 2022 and compared the results to the actual data. We repeated the process by training on 2022 and predicting 2023, and then by training on both 2021 and 2022 to predict 2023. In each iteration of this experiment, the R^2 between the real and predicted values was greater than 0.86, meaning the model explained more than 86% of the variability.

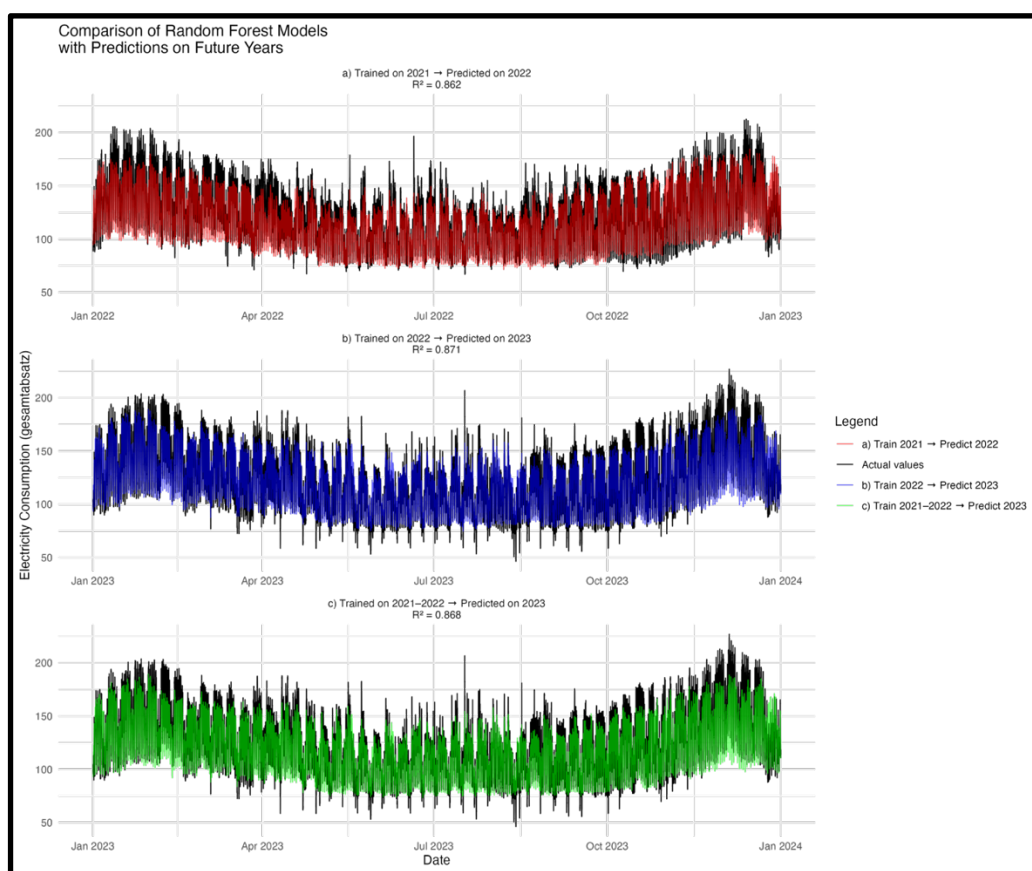


Figure 13 Graphs about the experience.

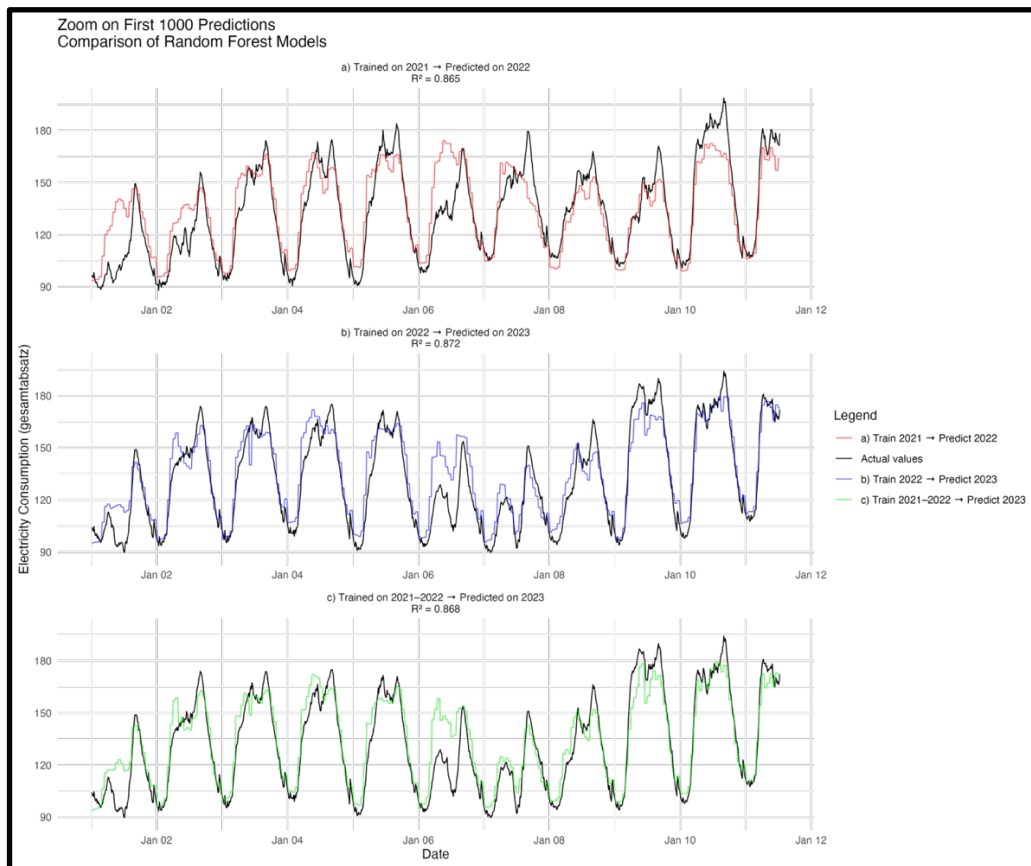


Figure 14 Graphs about the experience (zoom)

4. Can We Do Better with XGBoost?

An alternative to Random Forest is XGBoost. It is a machine learning model that is more complex to configure but potentially more accurate.

Automatic Optimization of an XGBoost Model

To find the best possible parameters for our model, we used an algorithm that automatically searches for optimal settings.

This was done using a random hyperparameter search method via the *caret* package. The method tests a number of randomly chosen parameter combinations and evaluates each model through cross-validation.

Once the best parameters were identified, the final model was trained on the full dataset. Its performance was then measured using the R^2 coefficient, which indicates the prediction accuracy.

The R^2 on the training data was over 0.98, similar to the Random Forest model.

The optimal parameters determined by the algorithm are as follows:

```
modele_xgb <- xgboost(data = X_matrix, label = y,  
                      nrounds = 844, max_depth = 8, eta = 0.378,  
                      gamma = 2.83, colsample_bytree = 0.648,  
                      min_child_weight = 5, subsample = 0.993,  
                      objective = "reg:squarederror", verbose = 0)
```

Figure 15 XGBoost parameters

Here is a breakdown of each parameter to better understand its importance:

- `nrounds = 844`: Number of training iterations (i.e., successive trees). A higher number allows the model to learn more, but increases the risk of overfitting.
- `max_depth = 8`: Maximum depth of the trees. Deeper trees can model more complex relationships, but also raise the risk of overfitting.
- `eta = 0.378`: Learning rate. It controls how quickly the model updates during training. A moderate value (like this one) provides a good balance between speed and accuracy.
- `gamma = 2.83`: Minimum loss reduction required to make a further split. A higher value increases regularization by preventing unnecessary splits, which helps limit complexity.
- `colsample_bytree = 0.648`: Fraction of columns (features) used for each tree. Here, about 65% of the variables are used, which promotes diversity among the trees.
- `min_child_weight = 5`: Minimum sum of instance weight (Hessian) needed in a child node. This avoids creating branches based on very small subsets and helps prevent overfitting.
- `subsample = 0.993`: Fraction of training data used for each tree. In this case, almost all the data are used each time, which favors model stability.
- `objective = "reg:squarederror"`: The objective function for regression, which minimizes the mean squared error. This is the standard setting for numeric prediction tasks.

These parameters make it possible to build a powerful but well-regularized model. They ensure a good balance between complexity, performance, and generalization, while reducing the risk of the model overfitting the training data.

Python: A Faster Language than R for Machine Learning

To generate the new predictions, we chose to use Python, as machine learning libraries in Python are, according to several sources, more optimized than those in R. This was also an opportunity to try a different language and bring more diversity to the project.

With Python, the XGBoost prediction program runs almost instantly. We kept the previously determined parameters and repeated the same experiment as with Random Forest: training the model on one year of data, then making predictions for the following year and comparing them to the actual values.

```
# Modèle avec paramètres fixes
fixed_model = XGBRegressor(
    objective="reg:squarederror",
    n_estimators=844,
    max_depth=8,
    learning_rate=0.378,
    gamma=2.83,
    colsample_bytree=0.648,
    min_child_weight=5,
    subsample=0.993,
    verbosity=0
)
```

Figure 16 XGBoost model with optimal parameters

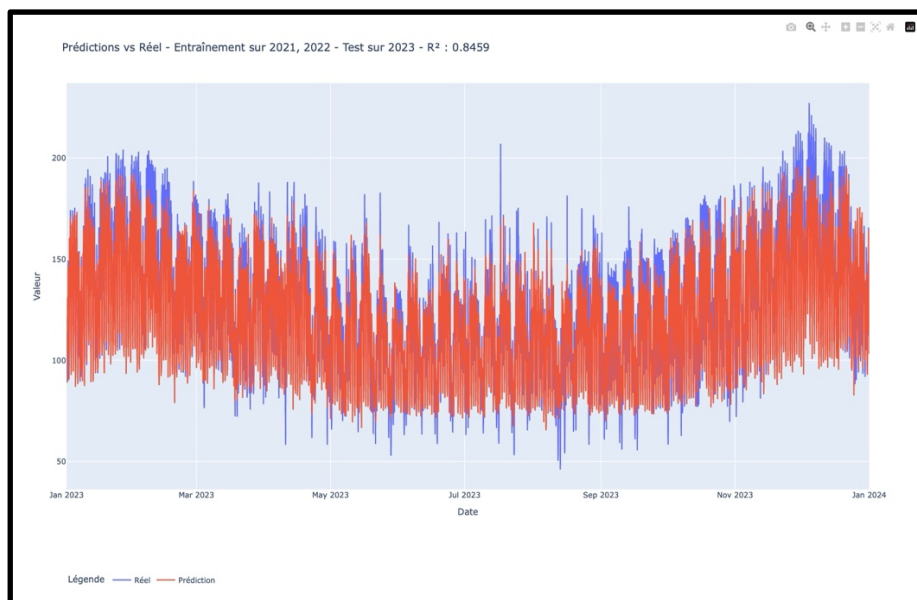


Figure 17 Comparison between the prediction and the reel values

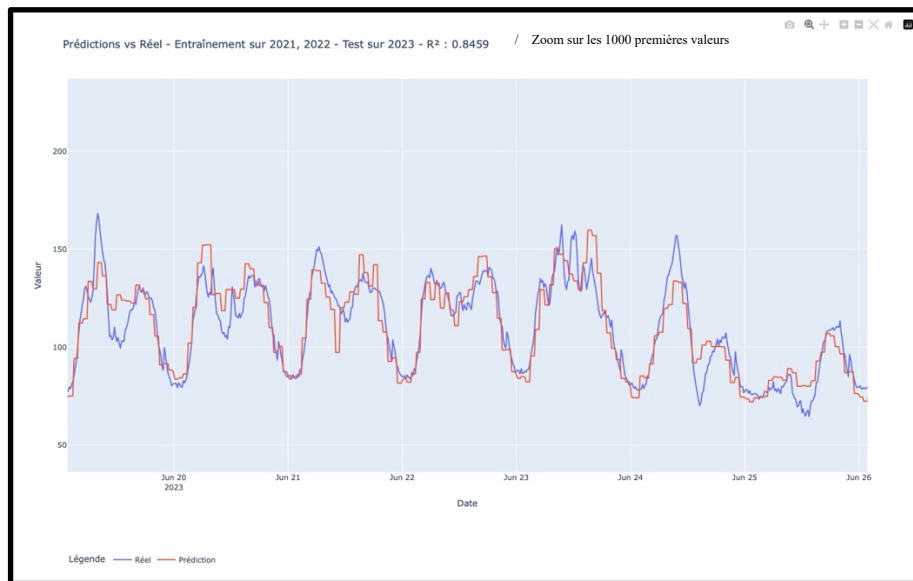


Figure 18 Comparison between the prediction and the reel values (zoom)

The results of this experiment are excellent: we obtained R^2 values ranging between 0.84 and 0.85 between the predictions and the training data.

V – Conclusion

This project allowed me to apply all the skills I acquired during my training, especially in data processing, exploratory analysis, and predictive modeling. Starting from a rich but complex dataset, I was able to extract relevant insights, identify the main factors influencing electricity consumption, and test various modeling approaches—from linear regression to more advanced methods like Random Forest and XGBoost.

Among the models tested, Random Forest proved to be the most robust for predicting Linz AG's electricity consumption, achieving an R^2 above 0.86 in generalization.

Beyond the technical aspect, this project also made me more aware of the importance of weather and time-related data in energy systems, as well as the industrial challenges linked to demand forecasting. It represents a significant step in my journey toward becoming a Data Scientist.

Second Project: Biostatistical Study on Yeast DNA Evolution

I – Introduction

As part of my second assignment in the Department of Statistics at JKU, I was tasked with reproducing and interpreting graphical results from the scientific study “*Crossing design shapes patterns of genetic variation in synthetic recombinant populations of *Saccharomyces cerevisiae**”, published in 2021 by Phillips et al. This study explores how different crossing strategies affect genetic diversity within synthetic recombinant yeast populations.

The researchers compared two methods for constructing genetically diverse yeast populations: a simple mixing method (“K-type”) and a more structured approach involving pairwise crossings (“S-type”). The main goal was to determine which of these methods best maximizes initial genetic variation and maintains it over time—a key factor in experimental evolution studies (known as Evolve and Resequence, or E&R).

My task was to reproduce the study’s key figures, including SNP frequency spectra, levels of heterozygosity and haplotype diversity, as well as potential signatures of selection over successive reproduction cycles. These visualizations are supported by a summary of the experimental protocol, the indicators used, and their relevance for analyzing genetic evolution in a controlled setting.

At the time of writing this report, the project is still in progress. Below is the most recent stage of my work.

II – Understanding and Summary of the Study

The researchers focused on how to build yeast populations (*Saccharomyces cerevisiae*) for studying their evolution in a lab setting. To ensure reliable results, these populations must have good genetic diversity from the start.

They compared two methods for combining yeast strains:

- A simple method where all strains are mixed at once (called the K method).
- A more careful method using pairwise crossings (S method), similar to arranging “marriages” between strains.

They tested both methods using 4, 8, or 12 different strains and followed how the populations evolved over several reproductive cycles.

What they found:

- The structured method (S) led to populations with richer genetic diversity and a more balanced distribution of strains.
- The simple method (K), especially with 12 strains, created unbalanced populations where some strains nearly disappeared.
- These differences were visible from the beginning, likely because some strains reproduced more successfully than others.
- The researchers also observed that certain regions of the genome changed more than others, which could suggest adaptation to the forced sexual reproduction process.
- However, traits like growth rate or reproduction capacity didn’t change much—except in one extreme case (K12).

Conclusion:

To properly observe yeast evolution in the lab, it’s better to take the time to do organized crossings. This results in more balanced and diverse populations, which are more useful for studying evolutionary processes.

III – My work

After spending time trying to understand the main points of the statistical study, I attempted to reproduce some of the graphs using the R programming language.

Here they are, along with the code used to generate them:

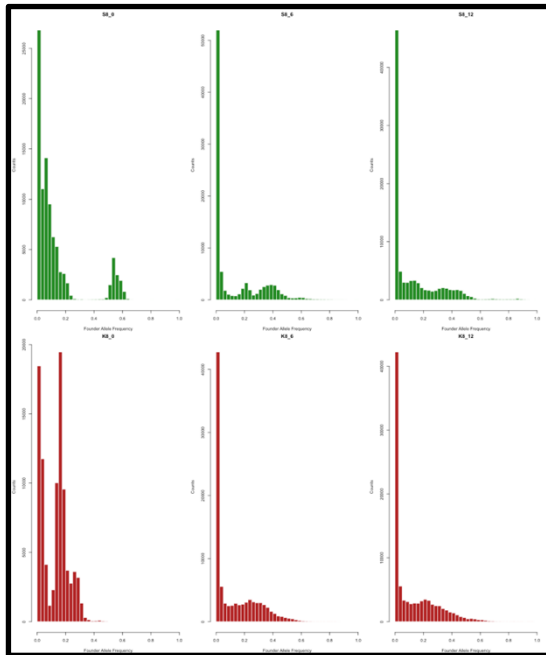


Figure 19 Allele Frequency Distribution for Population Size 4

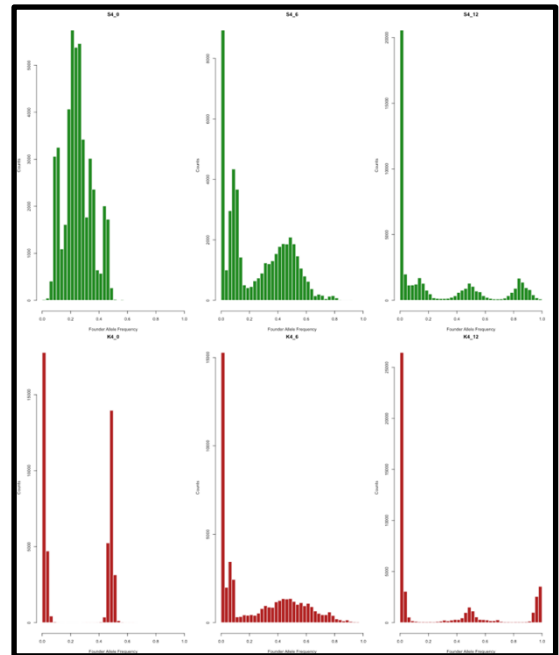


Figure 20 Allele Frequency Distribution for Population Size 8

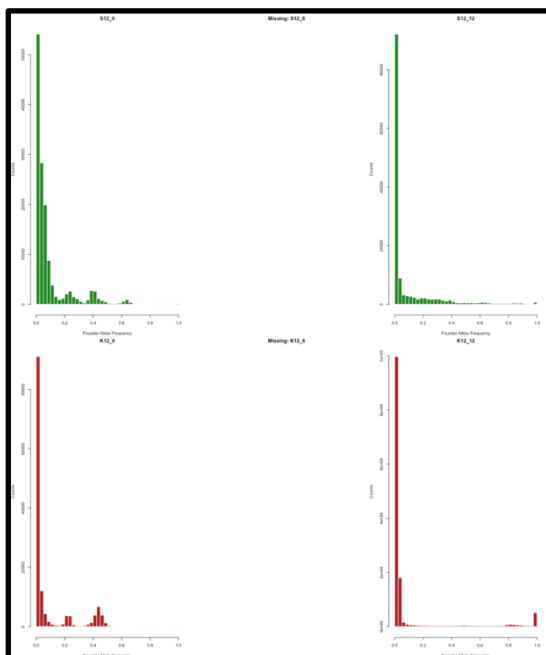


Figure 21 Allele Frequency Distribution for Population Size 12

Explanation of the R Code:

```
1 setwd("/Users/remierron/Library/CloudStorage/OneDrive-UniversitéPoitiers/biostats/dat
2
3
4 # ----- Fonctions -----
5
6 extract_frequencies <- function(df) {
7   freqs_raw <- strsplit(df$founderfreqs, ";")
8   freqs <- as.numeric(unlist(freqs_raw))
9   return(pmax(0, pmin(1, freqs)))
10 }
11
12 plot_snp_spectrum <- function(freqs, title, col = "steelblue") {
13   hist(freqs,
14     breaks = seq(0, 1, length.out = 41),
15     col = col,
16     include.lowest = TRUE,
17     main = title,
18     xlab = "Founder Allele Frequency",
19     ylab = "Counts",
20     border = "white")
21 }
22
23 # ----- Paramètres -----
24
25 types <- c("S", "K")
26 cycles <- c("0", "6", "12")
27 colors <- list("S" = "Forestgreen", "K" = "Firebrick")
28
29 # ----- Séparé par taille de fondation -----
30
31 for (size in c(4, 8, 12)) {
32   output_file <- paste0("SFS_founder", size, ".png")
33   png(output_file, width = 1000, height = 1200)
34   par(mfrow = c(2, 3), mar = c(4, 4, 2, 1)) # 2 types x 3 temps
35
36   for (type in types) {
37     for (cycle in cycles) {
38       label <- paste0(type, size, "_", cycle)
39       filename <- paste0(size, tolower(type), "_haps_", cycle, ".txt")
40
41       if (file.exists(filename)) {
42         df <- read.table(filename, header = TRUE, sep = "\t", stringsAsFactors = FALSE)
43         freqs <- extract_frequencies(df)
44         plot_snp_spectrum(freqs, title = label, col = colors[[type]])
45       } else {
46         plot.new()
47         title(main = paste("Missing:", label))
48       }
49     }
50   }
51
52   dev.off()
53   cat("Image générée :", output_file, "\n")
54 }
55
56 }
```

Figure 22 R script

For each population size, the script creates a PNG image containing six graphs (2 types $\times$ 3 time points). It reads the corresponding data files for each combination, extracts the frequencies, and plots the histogram using a specific color depending on the treatment type. If a file is missing, a blank graph is displayed with the label “Missing.”

These plots allow us to visualize how alleles are distributed among the founders and to compare this distribution across different experimental conditions.

This R script generates graphs showing the distribution of allele frequencies in the founder individuals (SFS: Site Frequency Spectrum), for different population sizes (4, 8, and 12 individuals), at three points in the evolutionary process (cycles 0, 6, and 12), and under two treatment types (S and K).

Two main functions are used: the first one, `extract_frequencies`, extracts allele frequencies from the data files, and the second one, `plot_snp_spectrum`, plots a histogram of the frequencies.

The data files are in .txt format and have a structure that is very difficult to work with. An example is included in the appendix.

IV – Conclusion of the Second Project

This second project allowed me to explore a real-world application of statistics in the field of evolutionary genetics. By working on reproducing the results from the study by Phillips et al. (2021), I not only deepened my understanding of key concepts such as genetic diversity, allele frequency, and selection, but also developed greater independence in interpreting complex scientific papers in English.

Creating the visualizations with R exposed me to real—and sometimes imperfect—biological datasets, which required careful attention in terms of processing, structuring, and representing the data. This experience also made me aware of how experimental choices in biology (especially crossing methods) can strongly influence results.

Additional Activities

I – Attending Bachelor’s and Master’s Courses in German and English

As part of my internship, I had the opportunity to attend several courses offered at JKU, both at the bachelor’s level (in German) and the master’s level (in English). I initially tried to follow an R programming course taught in German, but I quickly gave up on that idea because the language level required was beyond my current skills, which made it too difficult to understand.

On the other hand, I attended master’s courses taught in English, especially on experimental design and statistics. These courses were interesting in terms of content, but they were challenging to follow since I joined them in the middle of the semester. I was able to grasp the main ideas, but it was difficult to fully absorb everything, especially the more advanced theoretical concepts that were introduced without much background.

II – Seminar

I also attended a scientific seminar presented by Corinna Perchtold, a PhD student in the department, titled *Spatio-temporal Precipitation Modeling in Austria*. The presentation focused on modeling precipitation in Austria using spatio-temporal generalized additive models, with a focus on climate change. Three scenarios were analyzed: the monthly average precipitation, monthly maximum precipitation, and the maximum drought duration per month. Each was modeled using suitable statistical distributions (gamma, mixed GEV, negative binomial).

This seminar helped broaden my scientific knowledge and introduced me to a field that has not yet been widely covered in my studies.

III – Language and Cultural Gains

From a language perspective, this internship greatly improved my use of English—through daily conversations with colleagues, attending classes, and reading scientific articles. My spoken and written fluency improved significantly.

Culturally, I had the chance to visit several Austrian cities (Linz, Vienna, Steyr, Strobl, and others). These moments helped me integrate more easily while offering a true immersion in Austrian culture.



Figure 23 Point of view in Steyr

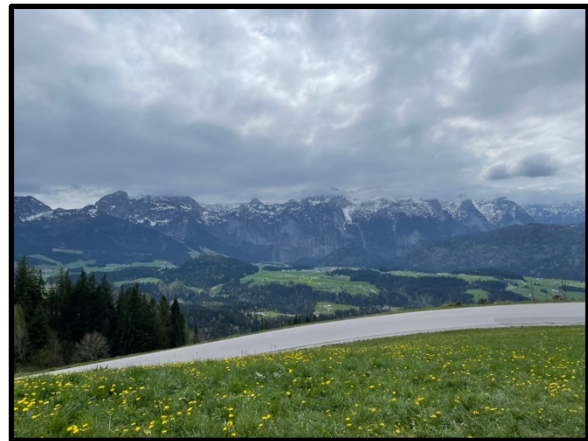


Figure 24 Point of view in the mountains



Figure 26 Cathedral of Wien



Figure 25 Botanic garden in Linz

Final Reflection

I – Professional Experience

This internship in the Department of Statistics at Johannes Kepler University Linz was a highly valuable professional experience. It allowed me to strengthen my technical skills in data science and acquire new ones, particularly in machine learning—a field I was not very familiar with before. I learned to use advanced methods such as Random Forest and XGBoost on my own, exploring how they work, how to tune their parameters, and how to evaluate their performance, both in R and Python.

The first project, focused on electricity consumption forecasting, was especially motivating. It allowed me to apply a complete analytical approach—from data preparation to the development of a robust predictive model. This work deepened my interest in applying data science to real-world energy challenges and encouraged me to pursue a master's degree with the goal of becoming a data scientist.

The second project, although still in progress, introduced me to another area of application: biostatistics. I learned how to interpret a complex scientific paper and reproduce some of its visualizations using R. While I still have a lot to learn, this experience gave me a stimulating introduction to the connections between statistics, biology, and genetics.

Finally, attending courses and seminars—even partially—helped broaden my knowledge in various fields and gave me the opportunity to experience a demanding and international academic environment.

II – Personal Experience

On a personal level, this internship presented a real language challenge. Attending courses in English, talking with colleagues, and understanding a scientific seminar required significant effort, especially at the beginning. However, I kept pushing myself to improve, and over time my English-both spoken and written-improved significantly.

Beyond language skills, this experience abroad helped me develop autonomy, adaptability, and cultural curiosity. I had the chance to explore several Austrian cities, connect with international students, and share enjoyable moments with my colleagues, which greatly broadened my perspective.

III – Final Conclusion

In summary, this internship at Johannes Kepler University Linz was a complete experience-both demanding and enriching. It allowed me to build strong technical skills in data science, explore new areas such as machine learning and biostatistics, and adapt to a stimulating international environment. Despite some challenges, especially with the language, I was able to adapt, grow, and get the most out of the experience.

This internship represents a turning point in my academic journey. It confirmed my strong interest in predictive modeling and reinforced my motivation to pursue a master's degree in order to become a Data Scientist. It was a key step that helped me gain confidence, independence, and clarity about my professional future.

Glossary

Haplotype diversity: A measure of genetic diversity based on the combination of genes inherited from a single parent.

Allele frequency: The proportion of a specific allele relative to all alleles of a gene in a population.

Heterozygosity: A measure of genetic diversity, corresponding to the presence of different alleles at a specific locus.

Hyperparameters: Parameters that control the behavior of a machine learning algorithm and are set before training begins.

JKU: Johannes Kepler University.

Linz AG: Company that manages electricity in the city of Linz.

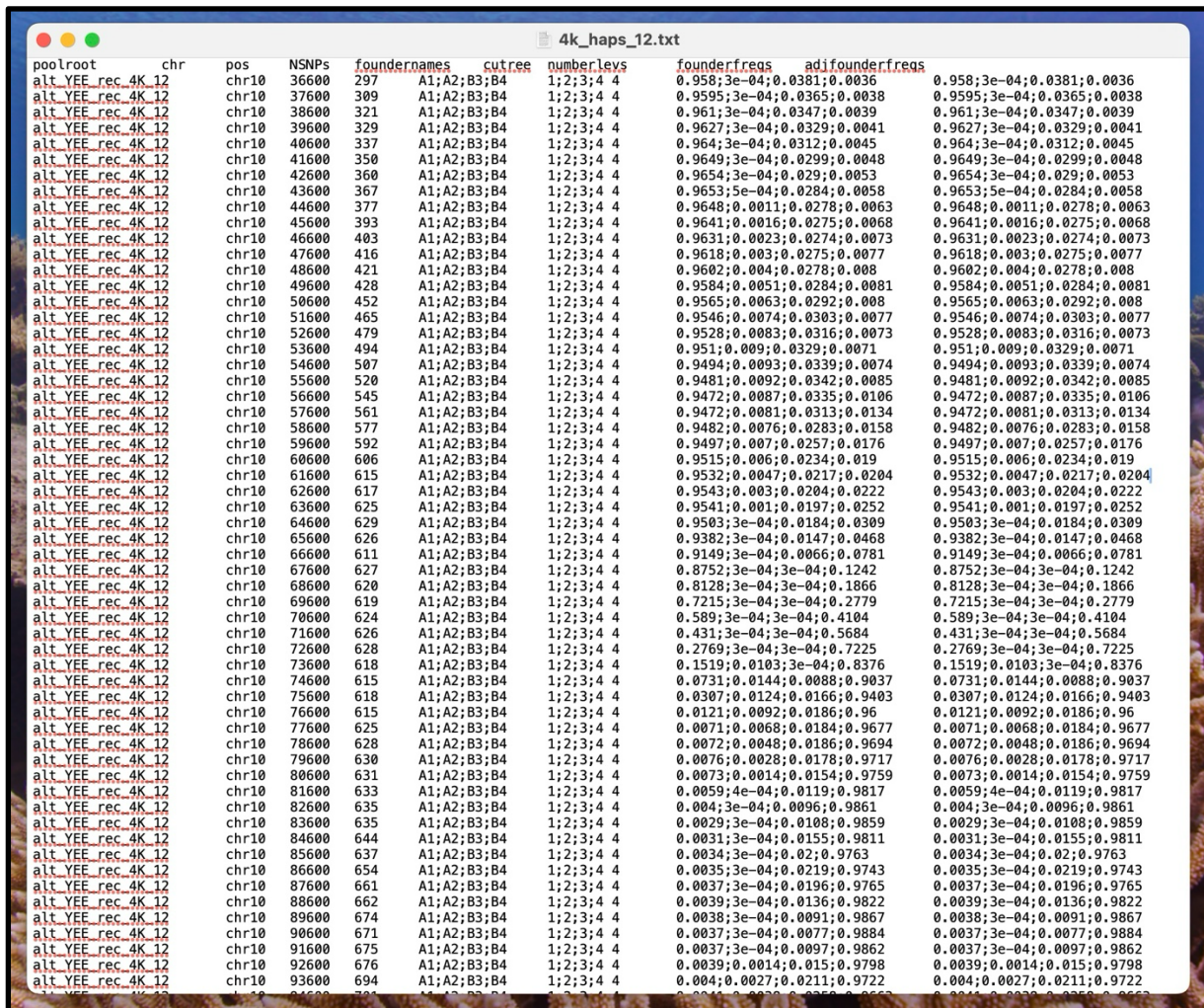
R²: A metric that measures the strength of the relationship between two variables. The closer it is to 1, the stronger the relationship.

Random Forest, XGBoost: Machine learning algorithms.

SNP (Single Nucleotide Polymorphism): A variation in a single nucleotide in the DNA sequence among individuals.

Overfitting: When a model learns the training data too well, at the expense of its ability to generalize to new data.

Appendix



poolroot	chr	pos	NSNPs	foundernames	cutree	numberleys	founderfreqs	adifounderfreqs
alt_YEE_rec_4K_12	chr10	36600	297	A1;A2;B3;B4	1;2;3;4	4	0.958;3e-04;0.0381;0.0036	0.958;3e-04;0.0381;0.0036
alt_YEE_rec_4K_12	chr10	37600	309	A1;A2;B3;B4	1;2;3;4	4	0.9595;3e-04;0.0365;0.0038	0.9595;3e-04;0.0365;0.0038
alt_YEE_rec_4K_12	chr10	38600	321	A1;A2;B3;B4	1;2;3;4	4	0.961;3e-04;0.0347;0.0039	0.961;3e-04;0.0347;0.0039
alt_YEE_rec_4K_12	chr10	39600	329	A1;A2;B3;B4	1;2;3;4	4	0.9627;3e-04;0.0329;0.0041	0.9627;3e-04;0.0329;0.0041
alt_YEE_rec_4K_12	chr10	40600	337	A1;A2;B3;B4	1;2;3;4	4	0.964;3e-04;0.0312;0.0045	0.964;3e-04;0.0312;0.0045
alt_YEE_rec_4K_12	chr10	41600	350	A1;A2;B3;B4	1;2;3;4	4	0.9649;3e-04;0.0299;0.0048	0.9649;3e-04;0.0299;0.0048
alt_YEE_rec_4K_12	chr10	42600	360	A1;A2;B3;B4	1;2;3;4	4	0.9654;3e-04;0.029;0.0053	0.9654;3e-04;0.029;0.0053
alt_YEE_rec_4K_12	chr10	43600	367	A1;A2;B3;B4	1;2;3;4	4	0.9653;5e-04;0.0284;0.0058	0.9653;5e-04;0.0284;0.0058
alt_YEE_rec_4K_12	chr10	44600	377	A1;A2;B3;B4	1;2;3;4	4	0.9648;0.0011;0.0278;0.0063	0.9648;0.0011;0.0278;0.0063
alt_YEE_rec_4K_12	chr10	45600	393	A1;A2;B3;B4	1;2;3;4	4	0.9641;0.0016;0.0275;0.0068	0.9641;0.0016;0.0275;0.0068
alt_YEE_rec_4K_12	chr10	46600	403	A1;A2;B3;B4	1;2;3;4	4	0.9631;0.0023;0.0274;0.0073	0.9631;0.0023;0.0274;0.0073
alt_YEE_rec_4K_12	chr10	47600	416	A1;A2;B3;B4	1;2;3;4	4	0.9618;0.003;0.0275;0.0077	0.9618;0.003;0.0275;0.0077
alt_YEE_rec_4K_12	chr10	48600	421	A1;A2;B3;B4	1;2;3;4	4	0.9602;0.004;0.0278;0.008	0.9602;0.004;0.0278;0.008
alt_YEE_rec_4K_12	chr10	49600	428	A1;A2;B3;B4	1;2;3;4	4	0.9584;0.0051;0.0284;0.0081	0.9584;0.0051;0.0284;0.0081
alt_YEE_rec_4K_12	chr10	50600	452	A1;A2;B3;B4	1;2;3;4	4	0.9565;0.0063;0.0292;0.008	0.9565;0.0063;0.0292;0.008
alt_YEE_rec_4K_12	chr10	51600	465	A1;A2;B3;B4	1;2;3;4	4	0.9546;0.0074;0.0303;0.0077	0.9546;0.0074;0.0303;0.0077
alt_YEE_rec_4K_12	chr10	52600	479	A1;A2;B3;B4	1;2;3;4	4	0.9528;0.0083;0.0316;0.0073	0.9528;0.0083;0.0316;0.0073
alt_YEE_rec_4K_12	chr10	53600	494	A1;A2;B3;B4	1;2;3;4	4	0.951;0.009;0.0329;0.0071	0.951;0.009;0.0329;0.0071
alt_YEE_rec_4K_12	chr10	54600	507	A1;A2;B3;B4	1;2;3;4	4	0.9494;0.0093;0.0339;0.0074	0.9494;0.0093;0.0339;0.0074
alt_YEE_rec_4K_12	chr10	55600	520	A1;A2;B3;B4	1;2;3;4	4	0.9481;0.0092;0.0342;0.0085	0.9481;0.0092;0.0342;0.0085
alt_YEE_rec_4K_12	chr10	56600	545	A1;A2;B3;B4	1;2;3;4	4	0.9472;0.0087;0.0335;0.0106	0.9472;0.0087;0.0335;0.0106
alt_YEE_rec_4K_12	chr10	57600	561	A1;A2;B3;B4	1;2;3;4	4	0.9472;0.0081;0.0313;0.0134	0.9472;0.0081;0.0313;0.0134
alt_YEE_rec_4K_12	chr10	58600	577	A1;A2;B3;B4	1;2;3;4	4	0.9482;0.0076;0.0283;0.0158	0.9482;0.0076;0.0283;0.0158
alt_YEE_rec_4K_12	chr10	59600	592	A1;A2;B3;B4	1;2;3;4	4	0.9497;0.007;0.0257;0.0176	0.9497;0.007;0.0257;0.0176
alt_YEE_rec_4K_12	chr10	60600	606	A1;A2;B3;B4	1;2;3;4	4	0.9515;0.006;0.0234;0.019	0.9515;0.006;0.0234;0.019
alt_YEE_rec_4K_12	chr10	61600	615	A1;A2;B3;B4	1;2;3;4	4	0.9532;0.0047;0.0217;0.0204	0.9532;0.0047;0.0217;0.0204
alt_YEE_rec_4K_12	chr10	62600	617	A1;A2;B3;B4	1;2;3;4	4	0.9543;0.003;0.0204;0.0222	0.9543;0.003;0.0204;0.0222
alt_YEE_rec_4K_12	chr10	63600	625	A1;A2;B3;B4	1;2;3;4	4	0.9541;0.001;0.0197;0.0252	0.9541;0.001;0.0197;0.0252
alt_YEE_rec_4K_12	chr10	64600	629	A1;A2;B3;B4	1;2;3;4	4	0.9503;3e-04;0.0184;0.0309	0.9503;3e-04;0.0184;0.0309
alt_YEE_rec_4K_12	chr10	65600	626	A1;A2;B3;B4	1;2;3;4	4	0.9382;3e-04;0.0147;0.0468	0.9382;3e-04;0.0147;0.0468
alt_YEE_rec_4K_12	chr10	66600	611	A1;A2;B3;B4	1;2;3;4	4	0.9149;3e-04;0.0066;0.0781	0.9149;3e-04;0.0066;0.0781
alt_YEE_rec_4K_12	chr10	67600	627	A1;A2;B3;B4	1;2;3;4	4	0.8752;3e-04;3e-04;0.1242	0.8752;3e-04;3e-04;0.1242
alt_YEE_rec_4K_12	chr10	68600	620	A1;A2;B3;B4	1;2;3;4	4	0.8128;3e-04;3e-04;0.1866	0.8128;3e-04;3e-04;0.1866
alt_YEE_rec_4K_12	chr10	69600	619	A1;A2;B3;B4	1;2;3;4	4	0.7215;3e-04;3e-04;0.2779	0.7215;3e-04;3e-04;0.2779
alt_YEE_rec_4K_12	chr10	70600	624	A1;A2;B3;B4	1;2;3;4	4	0.589;3e-04;3e-04;0.4104	0.589;3e-04;3e-04;0.4104
alt_YEE_rec_4K_12	chr10	71600	626	A1;A2;B3;B4	1;2;3;4	4	0.431;3e-04;3e-04;0.5684	0.431;3e-04;3e-04;0.5684
alt_YEE_rec_4K_12	chr10	72600	628	A1;A2;B3;B4	1;2;3;4	4	0.2769;3e-04;3e-04;0.7225	0.2769;3e-04;3e-04;0.7225
alt_YEE_rec_4K_12	chr10	73600	618	A1;A2;B3;B4	1;2;3;4	4	0.1519;0.0103;3e-04;0.8376	0.1519;0.0103;3e-04;0.8376
alt_YEE_rec_4K_12	chr10	74600	615	A1;A2;B3;B4	1;2;3;4	4	0.0731;0.0144;0.0088;0.9037	0.0731;0.0144;0.0088;0.9037
alt_YEE_rec_4K_12	chr10	75600	618	A1;A2;B3;B4	1;2;3;4	4	0.0307;0.0124;0.0166;0.9403	0.0307;0.0124;0.0166;0.9403
alt_YEE_rec_4K_12	chr10	76600	615	A1;A2;B3;B4	1;2;3;4	4	0.0121;0.0092;0.0186;0.96	0.0121;0.0092;0.0186;0.96
alt_YEE_rec_4K_12	chr10	77600	625	A1;A2;B3;B4	1;2;3;4	4	0.0071;0.0068;0.0184;0.9677	0.0071;0.0068;0.0184;0.9677
alt_YEE_rec_4K_12	chr10	78600	628	A1;A2;B3;B4	1;2;3;4	4	0.0072;0.0048;0.0186;0.9694	0.0072;0.0048;0.0186;0.9694
alt_YEE_rec_4K_12	chr10	79600	630	A1;A2;B3;B4	1;2;3;4	4	0.0076;0.0028;0.0178;0.9717	0.0076;0.0028;0.0178;0.9717
alt_YEE_rec_4K_12	chr10	80600	631	A1;A2;B3;B4	1;2;3;4	4	0.0073;0.0014;0.0154;0.9759	0.0073;0.0014;0.0154;0.9759
alt_YEE_rec_4K_12	chr10	81600	633	A1;A2;B3;B4	1;2;3;4	4	0.0059;4e-04;0.0119;0.9817	0.0059;4e-04;0.0119;0.9817
alt_YEE_rec_4K_12	chr10	82600	635	A1;A2;B3;B4	1;2;3;4	4	0.004;3e-04;0.0096;0.9861	0.004;3e-04;0.0096;0.9861
alt_YEE_rec_4K_12	chr10	83600	635	A1;A2;B3;B4	1;2;3;4	4	0.0029;3e-04;0.0108;0.9859	0.0029;3e-04;0.0108;0.9859
alt_YEE_rec_4K_12	chr10	84600	644	A1;A2;B3;B4	1;2;3;4	4	0.0031;3e-04;0.0155;0.9811	0.0031;3e-04;0.0155;0.9811
alt_YEE_rec_4K_12	chr10	85600	637	A1;A2;B3;B4	1;2;3;4	4	0.0034;3e-04;0.02;0.9763	0.0034;3e-04;0.02;0.9763
alt_YEE_rec_4K_12	chr10	86600	654	A1;A2;B3;B4	1;2;3;4	4	0.0035;3e-04;0.0219;0.9743	0.0035;3e-04;0.0219;0.9743
alt_YEE_rec_4K_12	chr10	87600	661	A1;A2;B3;B4	1;2;3;4	4	0.0037;3e-04;0.0196;0.9765	0.0037;3e-04;0.0196;0.9765
alt_YEE_rec_4K_12	chr10	88600	662	A1;A2;B3;B4	1;2;3;4	4	0.0039;3e-04;0.0136;0.9822	0.0039;3e-04;0.0136;0.9822
alt_YEE_rec_4K_12	chr10	89600	674	A1;A2;B3;B4	1;2;3;4	4	0.0038;3e-04;0.0091;0.9867	0.0038;3e-04;0.0091;0.9867
alt_YEE_rec_4K_12	chr10	90600	671	A1;A2;B3;B4	1;2;3;4	4	0.0037;3e-04;0.0077;0.9884	0.0037;3e-04;0.0077;0.9884
alt_YEE_rec_4K_12	chr10	91600	675	A1;A2;B3;B4	1;2;3;4	4	0.0037;3e-04;0.0097;0.9862	0.0037;3e-04;0.0097;0.9862
alt_YEE_rec_4K_12	chr10	92600	676	A1;A2;B3;B4	1;2;3;4	4	0.0039;0.0014;0.015;0.9798	0.0039;0.0014;0.015;0.9798
alt_YEE_rec_4K_12	chr10	93600	694	A1;A2;B3;B4	1;2;3;4	4	0.004;0.0027;0.0211;0.9722	0.004;0.0027;0.0211;0.9722

Figure 27 Exemple of a file used in the biostatistic project

As part of my undergraduate degree in **Data Science (BUT)**, I completed a twelve-week internship in the Department of Applied Statistics at **Johannes Kepler University Linz**, in Austria. This report summarizes all the work I carried out, as well as the technical, scientific, and interpersonal skills I developed during the experience.

I first worked on an **electricity consumption forecasting** project for **Linz AG**, as part of a university competition. This involved handling high-frequency energy and weather data, performing exploratory analyses, and building predictive models in both R and Python using advanced methods such as Random Forest and XGBoost.

At the same time, I was involved in a **biostatistics** project aimed at reproducing graphical results from a scientific study on the genetic evolution of yeast. This second part of the internship allowed me to deepen my understanding of statistical genetics and data visualization, while gaining autonomy in interpreting complex scientific publications.

The internship also offered a full linguistic and cultural immersion, strengthening my adaptability, scientific rigor, and openness to international experiences. It represents a key step in my academic journey, confirming my desire to pursue a master's degree in data science.